

Progress Report on a Minimal Bright–Dark Model for Nonlinear SOC Signatures in a Spin-Peierls Context

Abstract

This note summarizes the modeling work completed so far for a minimal bright–dark Liouville-space model motivated by nonlinear spectroscopy of CuGeO_3 -like spin chains. The goal is not yet to provide a full microscopic description of CuGeO_3 , but to define and validate a sequence of numerical controls capable of separating three possible origins of a nonlinear signal: constant bright–dark spin-orbit mixing, lattice or spin-Peierls dimerisation, and local spin-correlation effects. The present results show that a scalar renormalisation of bright–dark mixing is insufficient to distinguish dimerisation from local spin correlations. Adding an explicit dynamical lattice coordinate breaks this degeneracy, and independent temperature/field trends provide a practical diagnostic for identifying whether a spectral feature tracks the spin-Peierls order parameter or persists in the uniform chain regime.

1 Motivation

The original objective was to formulate a tractable model for extracting two-dimensional nonlinear spectra of an $S = 1/2$ spin-chain material such as CuGeO_3 . At low temperature, the dominant material physics is expected to be strongly tied to the spin-Peierls lattice instability rather than to a bare spin-orbit coupling (SOC) alone. The working question is therefore more specific: can nonlinear spectroscopy, especially when exciting d -band degrees of freedom, reveal a contribution associated with SOC-like bright–dark mixing while avoiding a false assignment of lattice-driven features to SOC?

The strategy has been to build a sequence of minimal controls. Each control adds only one conceptual ingredient, so that failure modes remain interpretable. The tests were implemented with the Liouville-space solver using third-order one-quantum pathways generated by the NQ protocol, with emphasis on rephasing spectra and finite-window integrals around diagonal and cross peaks.

2 Minimal orbital model

The common orbital sector contains a ground state $|g\rangle$, a dark excited state $|D\rangle$, and a bright excited state $|B\rangle$. The unperturbed Hamiltonian is

$$H_0 = \Delta_D |D\rangle\langle D| + \Delta_B |B\rangle\langle B|, \quad (1)$$

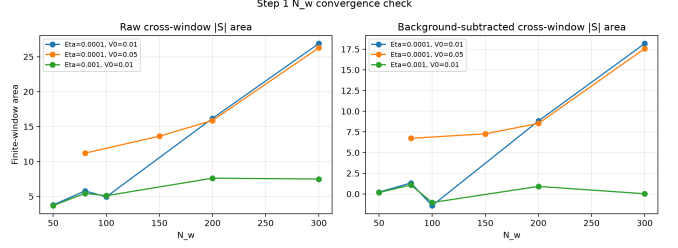


Figure 1: Step 1 convergence diagnostic. The cross-window finite integral depends on both linewidth and grid density. The model response is present, but later comparisons should be interpreted as controlled relative tests rather than as fully converged absolute intensities.

with the bright state carrying the optical dipole. Bright–dark mixing is introduced through

$$L_{BD} = |D\rangle\langle B| + |B\rangle\langle D|. \quad (2)$$

The simplest model uses

$$H_{\text{mix}} = V_{BD} L_{BD}. \quad (3)$$

For the validated runs summarized below, the representative orbital energies were $\Delta_D = 0.90$ eV and $\Delta_B = 1.00$ eV, with broadening parameter $\eta = 10^{-3}$ unless otherwise noted.

3 Step 1: minimal bright–dark validation

The first test established that the solver and the finite-window analysis can detect bright–dark mixing in a controlled two-level excited-state model. The control parameter was a static mixing amplitude V_0 , compared against a $V_0 = 0$ baseline. Rephasing spectra were generated for several values of the linewidth parameter η and frequency-grid size N_ω .

The important outcome is qualitative rather than a claim of perfect numerical convergence. The cross-window response is sensitive to V_0 , but the extracted finite-window integrals also depend on η and grid resolution, especially when narrow lines are sampled on a coarse grid. This means Step 1 is sufficient as a validation of the spectral machinery and pathway extraction, but not as a final convergence study.

4 Step 2: scalar separation of SOC, dimerisation, and spin correlations

The second step tested whether three scalar contributions to bright-dark mixing could be distinguished spectroscopically:

$$V_{BD}^{\text{eff}} = V_0 + \lambda_\delta \delta + \lambda_C C_1. \quad (4)$$

Here δ is the spin-Peierls/dimerisation order parameter, while $C_1 = \langle \mathbf{S}_i \cdot \mathbf{S}_{i+1} \rangle$ is a local spin correlation proxy. The distinction is physical: δ vanishes above the spin-Peierls transition, whereas C_1 can remain finite in the uniform chain.

Three cases were compared: constant SOC-like mixing, dimerisation-enhanced mixing, and spin-correlation-enhanced mixing. The dimerisation and spin-correlation cases were matched to the same $V_{BD}^{\text{eff}} = 0.02$, while the constant control used $V_{BD}^{\text{eff}} = 0.01$.

The result is decisive: the scalar model distinguishes weak and enhanced effective mixing, but it cannot identify the microscopic origin of the enhancement. Dimerisation and C_1 produce numerically identical spectra when they enter only through the same scalar V_{BD}^{eff} .

5 Step 3: dynamic lattice coordinate

Step 3 added a truncated harmonic coordinate representing a dynamical lattice degree of freedom. The mixing Hamiltonian became

$$H_{\text{mix}} = V_{BD}^{\text{static}} L_{BD} + g_Q Q L_{BD}, \quad (5)$$

with

$$Q = a + a^\dagger, \quad V_{BD}^{\text{static}} = V_0 + \lambda_\delta \delta + \lambda_C C_1. \quad (6)$$

The central question was whether adding the operator-valued term $g_Q Q L_{BD}$ creates spectral information that cannot be reproduced by a static scalar C_1 control.

The two static controls remained identical at matched $V_{BD}^{\text{static}} = 0.02$, reproducing the Step 2 degeneracy inside the larger Hilbert space. The dynamic dimerisation/phonon case, however, differed strongly from the static controls. The relative maximum full-spectrum difference between dynamic and static dimerisation was 5.46×10^{-1} .

Table 1: Step 2 finite-window metrics for the rephasing component.

Case	V_{BD}^{eff}	A_{cross}	diff.
Constant	0.01	10.210040	–
Dimer.	0.02	13.682378	0.657
C_1	0.02	13.682378	2.62×10^{-17}

Table 2: Step 3 rephasing finite-window metrics.

Case	g_Q	A_{cross}	cross/diag
Static C_1	0	14.160315	0.142697
Static δ	0	14.160315	0.142697
Dynamic δ	0.010	11.761513	0.149235

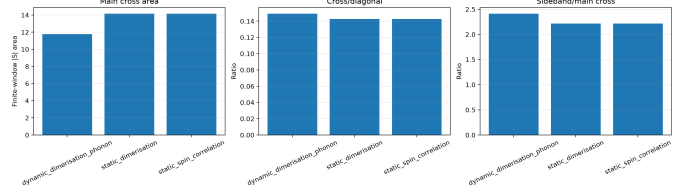


Figure 2: Step 3 aggregate metrics. The static dimerisation and static spin-correlation controls remain degenerate, while the dynamic phonon coordinate changes the finite-window response and sideband-related metrics.

The conclusion is that a lattice contribution must enter through a dynamical or sector-specific degree of freedom if it is to be distinguishable from a scalar local spin-correlation control.

6 Step 4: independent temperature and field trends

The fourth step introduced independent phenomenological trends for dimerisation and local spin correlations. The spin-Peierls transition temperature was modeled as

$$T_{\text{SP}}(B) = \max[0, T_{\text{SP},0}(1 - \alpha_B B^2)], \quad (7)$$

with

$$\delta(T, B) = \begin{cases} \delta_0 (1 - T/T_{\text{SP}}(B))^\beta, & T < T_{\text{SP}}(B), \\ 0, & T \geq T_{\text{SP}}(B). \end{cases} \quad (8)$$

The dynamic lattice coupling was scaled as

$$g_Q^{\text{eff}} = g_{Q0} \frac{\delta(T, B)}{\delta(T_{\text{ref}}, B_{\text{ref}})}. \quad (9)$$

In contrast, $C_1(T, B)$ was kept finite in both the dimerised and uniform regimes. This is the essential trend-level control: if C_1 were forced to zero with δ , one could not distinguish a spin-Peierls order parameter from a generic spin-correlation channel.

Table 3: Step 4 trend-level controls.

Case	δ	C_1	g_Q^{eff}	A_{cross}
Low- T lat.	0.007071	-0.378667	0.010	12.554241
High- T off	0	-0.329471	0	10.919127
High- B off	0	-0.346199	0	10.919127
High- T C_1	0	-0.329471	0	16.266623

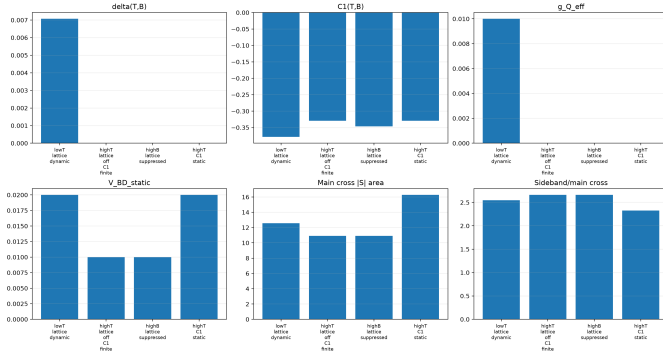


Figure 3: Step 4 trend metrics. The lattice channel follows $\delta(T, B)$ and turns off above the transition or under field suppression, while the C_1 -static control can remain strong even when $\delta = 0$.

The low-temperature lattice case differs strongly from the high-temperature lattice-off control, with relative maximum full-spectrum difference 6.76×10^{-1} . The high-field and high-temperature lattice-off controls are identical in the minimal Hamiltonian because both have $\delta = 0$, $g_Q^{\text{eff}} = 0$, and the same scalar mixing. The high-temperature C_1 static control remains large even with $\delta = 0$; its difference from the high-temperature lattice-off baseline is 5.98×10^{-1} .

7 Current interpretation

The completed tests support four working conclusions.

First, the minimal bright–dark model is adequate as a numerical validation platform. It produces controlled diagonal and cross-peak responses, but absolute finite-window intensities should be treated carefully because they depend on linewidth and grid density.

Second, a scalar effective mixing $V_{BD}^{\text{eff}} = V_0 + \lambda_\delta \delta + \lambda_C C_1$ is not enough to identify whether an enhanced signal is caused by spin-Peierls dimerisation or by local spin correlations. At matched scalar mixing, the spectra are identical.

Third, an explicit dynamical lattice coordinate provides a viable way to break that degeneracy. The operator-valued term $g_Q Q L_{BD}$ changes the nonlinear response even when the static bright–dark mixing is held fixed.

Fourth, the correct experimental or phenomenological diagnostic should be trend-based. A feature that follows $\delta(T, B)$ and vanishes above the spin-Peierls transition is compatible with lattice/spin-Peierls control. A feature that persists when $\delta = 0$ but C_1 remains finite cannot be assigned uniquely to the dimerised phase.

8 Limitations and next steps

The present model is intentionally minimal. It does not yet include a true spinon continuum, a rigorous momentum-

sector treatment, or a microscopic phonon bath. Momentum integration has been kept off in the validated trend tests in order to isolate the bright–dark and lattice controls. This is a reasonable staging choice, but it means that any comparison to a uniform spin-chain regime remains phenomenological.

The next useful calculations are:

1. Sweep T across T_{SP} and track the same finite-window observables used in Step 4.
2. Sweep B at fixed low temperature to test field suppression of the lattice channel.
3. Scan g_Q and ω_Q to identify which observables scale with the dynamic phonon coordinate rather than with scalar mixing.
4. Only after the above controls are stable, revisit momentum-sector coherence and possible spinon-inspired response functions.

9 Summary

The main modeling lesson is that “SOC sensitivity” cannot be inferred from a temperature-dependent nonlinear feature alone. In a spin-Peierls material, a low-temperature enhancement can arise from lattice dimerisation, from local spin correlations, or from genuine SOC-enabled bright–dark mixing. The tests completed so far define a practical hierarchy of controls: validate the minimal bright–dark response, show the failure of scalar separation, add a dynamic lattice coordinate, and finally compare independent temperature/field trends. This provides a concrete foundation for the next phase of the CuGeO_3 modeling effort without prematurely committing to a full microscopic spin-chain solver.